

Chapter 2: Getting to Know Your Data

- Data Objects and Attribute Types
- Basic Statistical Descriptions of Data
- Data Visualization
- Measuring Data Similarity and Dissimilarity
- Summary



Similarity and Dissimilarity

- **Similarity**

- Numerical measure of how alike two data objects are
- Value is higher when objects are more alike
- Often falls in the range $[0,1]$

- **Dissimilarity** (e.g., distance)

- Numerical measure of how different two data objects are
- Lower when objects are more alike
- Minimum dissimilarity is often 0
- Upper limit varies

- **Proximity** refers to a similarity or dissimilarity

Data Matrix and Dissimilarity Matrix

- Data matrix (*object-by-attribute structure*)

- n data points with p dimensions
- Two modes

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

- Dissimilarity matrix (*object-by-object structure*)

- n data points, but registers only the distance
- A triangular matrix
- Single mode

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ : & : & : & & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

Similarity and Dissimilarity

- where $d(i, j)$ is the measured **dissimilarity** or “difference” between objects i and j .
- $d(i, j)$ is a non-negative number that is close to 0 when objects i and j are highly similar or “near” each other, and becomes larger the more they differ.
- $d(i, i) = 0$; that is, the difference between an object and itself is 0.
- $d(i, j) = d(j, i)$.
- Measures of similarity can be expressed as a function of measures of dissimilarity.
- For example, for nominal data, $sim(i, j) = 1 - d(i, j)$
- where $sim(i, j)$ is the similarity between objects i and j .

Proximity Measure for Nominal Attributes

- Can take 2 or more states, e.g., red, yellow, blue, green (generalization of a binary attribute)
- Let the number of states of a nominal attribute be M .
- The states can be denoted by letters, symbols, or a set of integers, such as $1, 2, \dots, M$.
- The dissimilarity between two objects i and j can be computed based on the ratio of mismatches:

$$d(i, j) = \frac{p - m}{p}$$

- m : # of matches.
- p : total # of attributes.

Proximity Measure for Nominal Attributes

- sample data

Object Identifier	test-1 (nominal)
1	code A
2	code B
3	code C
4	code A

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

- We have one nominal attribute, *test-1*, we set $p = 1$ in the Equation. so that $d(i, j)$ evaluates to 0 if objects i and j match, and 1 if the objects differ.
- All objects are dissimilar except objects 1 and 4 (i.e., $d(4,1) = 0$).

Proximity Measure for Binary Attributes

- If all binary attributes are having the same weight, we have the 2 x 2 contingency table.

Contingency Table for Binary Attributes

		Object j		
		1	0	sum
Object i	1	q	r	$q + r$
	0	s	t	$s + t$
	sum	$q + s$	$r + t$	p

- q = # of attributes that equal 1 for both objects i and j
- r = # of attributes that equal 1 for object i but equal 0 for object j
- s = # of attributes that equal 0 for object i but equal 1 for object j
- t = # of attributes that equal 0 for both objects i and j .
- The total number of attributes is p , where $p = q + r + s + t$.

Proximity Measure for Binary Attributes

- Dissimilarity that is based on symmetric binary attributes is called **symmetric binary dissimilarity**.
- If objects i and j are described by symmetric binary attributes, then the dissimilarity between i and j is

$$d(i, j) = \frac{r + s}{q + r + s + t}.$$

- For asymmetric binary attributes the dissimilarity is called **asymmetric binary dissimilarity**.
- where the number of negative matches, t , is considered unimportant and is thus ignored in the following computation

$$d(i, j) = \frac{r + s}{q + r + s}.$$

Proximity Measure for Binary Attributes

- we can measure the difference between two binary attributes based on the notion of similarity instead of dissimilarity.
- For example, the **asymmetric binary similarity** between the objects i and j can be computed as

$$sim(i, j) = \frac{q}{q + r + s} = 1 - d(i, j).$$

- The coefficient $sim(i, j)$ of the above Equation is called the **Jaccard coefficient**

Dissimilarity between Binary Variables

■ Example

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

- Gender is a symmetric attribute
- The remaining attributes are asymmetric binary
- Let the values Y and P be 1, and the value N 0

$$d(jack, mary) = \frac{0 + 1}{2 + 0 + 1} = 0.33$$

$$d(jack, jim) = \frac{1 + 1}{1 + 1 + 1} = 0.67$$

$$d(jim, mary) = \frac{1 + 2}{1 + 1 + 2} = 0.75$$

Distance on Numeric Data

- Measures that are commonly used for computing the dissimilarity of objects described by numeric attributes.
- These measures include the
 - *Euclidean*
 - *Manhattan*
 - *Minkowski distances*.
- The most popular distance measure is **Euclidean distance**.
- Let $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ be two objects described by p numeric attributes.
- The Euclidean distance between objects i and j is defined as

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ip} - x_{jp})^2}.$$

Distance on Numeric Data

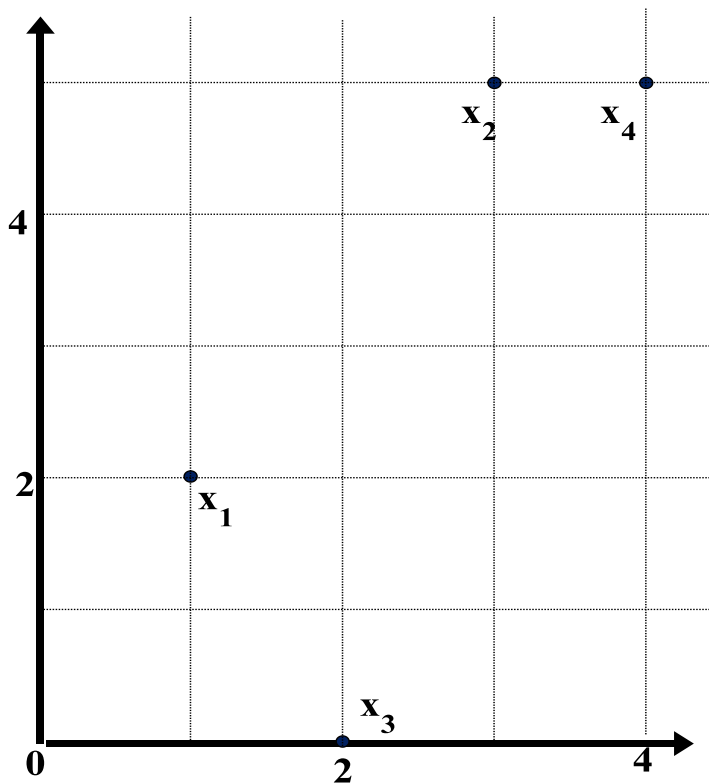
- **Manhattan distance** is defined as

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots + |x_{ip} - x_{jp}|.$$

- Mathematical properties of the Euclidean and the Manhattan distance
 - **Non-negativity:** $d(i, j) \geq 0$: Distance is a non-negative number.
 - **Identity of indiscernibles:** $d(i, i) = 0$: The distance of an object to itself is 0.
 - **Symmetry:** $d(i, j) = d(j, i)$: Distance is a symmetric function.
 - **Triangle inequality:** $d(i, j) \leq d(i, k) + d(k, j)$: Going directly from object i to object j in space is no more than making a detour over any other object k

Distance on Numeric Data

- Let $\mathbf{x1} = (1, 2)$ and $\mathbf{x2} = (3, 5)$ represent two objects
- The Euclidean distance is $\sqrt{2^2 + 3^2} = 3.61$.
- The Manhattan distance is $2 + 3 = 5$.



Data Matrix

point	attribute1	attribute2
$x1$	1	2
$x2$	3	5
$x3$	2	0
$x4$	4	5

Dissimilarity Matrix

- With Euclidean Distance

	<i>x1</i>	<i>x2</i>	<i>x3</i>	<i>x4</i>
<i>x1</i>	0			
<i>x2</i>	3.61	0		
<i>x3</i>	2.24	5.1	0	
<i>x4</i>	4.24	1	5.39	0

- With Manhattan Distance

L	<i>x1</i>	<i>x2</i>	<i>x3</i>	<i>x4</i>
<i>x1</i>	0			
<i>x2</i>	5	0		
<i>x3</i>	3	6	0	
<i>x4</i>	6	1	7	0

Distance on Numeric Data

- **Minkowski distance** is a generalization of the Euclidean and Manhattan distances.
- It is defined as

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h},$$

- where h is a real number such that $h \geq 1$.
- Such a distance is also called L_p **norm**.
- It represents the Manhattan distance when $h = 1$ (i.e., L_1 norm) and Euclidean distance when $h = 2$ (i.e., L_2 norm).

Distance on Numeric Data

- The **supremum distance** ($L_{\max}$ norm, L_{∞} norm) is a generalization of the Minkowski distance for $h \rightarrow \infty$.
- To compute it, we find the attribute f that gives the maximum difference in values between the two objects.
- This difference is the supremum distance, defined as:

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f |x_{if} - x_{jf}|.$$

- Let's use the same two objects, $\mathbf{x}_1 = (1, 2)$ and $\mathbf{x}_2 = (3, 5)$.
- The second attribute gives the greatest difference between values for the objects, which is $5 - 2 = 3$.
- This is the supremum distance between both objects.

Dissimilarity Matrix

point	attribute1	attribute2
$x1$	1	2
$x2$	3	5
$x3$	2	0
$x4$	4	5

▪ With Supremum Distance

L_{∞}	$x1$	$x2$	$x3$	$x4$
$x1$	0			
$x2$	3	0		
$x3$	2	5	0	
$x4$	3	1	5	0

Distance on Ordinal Data

- Ordinal attributes may also be obtained from the discretization of numeric attributes by splitting the value range into a finite number of categories.
- The range of a numeric attribute can be mapped to an ordinal attribute f having M_f states.
 - For example, the range of the interval-scaled attribute *temperature* can be organized into the following states: -30 to -10, -10 to 10, 10 to 30, representing the categories *cold temperature*, *moderate temperature*, and *warm temperature*, respectively.
- Let M represent the number of possible states that an ordinal attribute can have.
- These ordered states define the ranking $1, \dots, M_f$.

Distance on Ordinal Data

- f is an attribute from a set of ordinal attributes describing n objects.
- The dissimilarity computation with respect to f involves the following steps:

1. Replace each x_{if} by its corresponding rank,

$$r_{if} \in \{f_1, \dots, M_f\}.$$

2. Normalize the data by replacing the rank r_{if} of the i^{th} object in the f^{th} attribute by

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}.$$

3. Dissimilarity can then be computed using any of the distance measures.

Distance on Ordinal Data

- A Sample Data Table Containing Attributes of Mixed Type

Object Identifier	test-1 (nominal)	test-2 (ordinal)	test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

- Consider the *object-identifier* and the continuous ordinal attribute, *test-2*.
- There are three states for *test-2*: *fair*, *good*, and *excellent*, that is, $M_f = 3$.

Distance on Ordinal Data

- Step 1 - Replace each value for *test-2* by its rank, the four objects are assigned the ranks 3, 1, 2, and 3, respectively.
- Step 2 normalizes the ranking by mapping rank 1 to 0.0, rank 2 to 0.5, and rank 3 to 1.0.
- Step 3, we can use the Euclidean distance to for dissimilarity matrix:

$$\begin{bmatrix} 0 & 1.0 & 0.5 & 0 \\ 1.0 & 0 & 0.5 & 1.0 \\ 0.5 & 0.5 & 0 & 0.5 \\ 0 & 1.0 & 0.5 & 0 \end{bmatrix}.$$

- Therefore, objects 1 and 2 are the most dissimilar.

Attributes of Mixed Type

- A database may contain all attribute types
 - Nominal, symmetric binary, asymmetric binary, numeric, ordinal
- One may use a weighted formula to combine their effects

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

- The indicator $\delta_{ij}^{(f)} = 0$
 - If either (1) x_{if} or x_{jf} is missing
or (2) $x_{if} = x_{jf} = 0$ and attribute f is asymmetric binary
 - otherwise,

$$\delta_{ij}^{(f)} = 1.$$

Attributes of Mixed Type

- The contribution of attribute f to the dissimilarity between i and j is computed dependent on its type:
- f is numeric: use the normalized distance

$$d_{ij}^{(f)} = \frac{|x_{if} - x_{jf}|}{\max_h x_{hf} - \min_h x_{hf}}$$

- where h runs over all nonmissing objects for attribute f .
- f is binary or nominal:
 - $d_{ij}^{(f)} = 0$ if $x_{if} = x_{jf}$, or $d_{ij}^{(f)} = 1$ otherwise
- f is ordinal
 - Compute ranks r_{if} and $z_{if} = \frac{r_{if} - 1}{M_f - 1}$ treat z_{if} as numeric

dissimilarity matrices for each of the individual attributes

A Sample Data Table Containing Attributes of Mixed Type

Object Identifier	test-1 (nominal)	test-2 (ordinal)	test-3 (numeric)
1	code A	excellent	45
2	code B	fair	22
3	code C	good	64
4	code A	excellent	28

**Results of Test 1
(Nominal Attribute)**

$$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

**Results of Test 2
(Ordinal Attribute)**

$$\begin{bmatrix} 0 & & & \\ 1.0 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1.0 & 0.5 & 0 \end{bmatrix}.$$

Attributes of Mixed Type

- For numeric attributes,
 - let $\max_h x_h = 64$ and $\min_h x_h = 22$.
 - The difference between the two is used in the following equation to normalize the values of the dissimilarity matrix.
 - $d_{21} = (45 - 22) / (64 - 22)$. $d_{ij}^{(f)} = \frac{|x_{if} - x_{jf}|}{\max_h x_{hf} - \min_h x_{hf}}$
 - The resulting dissimilarity matrix for *test-3* is

$$\begin{bmatrix} 0 & & & \\ 0.55 & 0 & & \\ 0.45 & 1.00 & 0 & \\ 0.40 & 0.14 & 0.86 & 0 \end{bmatrix}$$

Attributes of Mixed Type

- We can now use the dissimilarity matrices for the three attributes in our computation
- The indicator $\delta_{ij}^{(f)} = 1$ for each of the three attributes, f
- We get, for example, $d(3, 1) = \frac{1(1) + 1(0.50) + 1(0.45)}{3} = 0.65$
- The resulting dissimilarity matrix obtained for the data described by the three attributes of mixed types is

$$\begin{bmatrix} 0 & & & \\ 0.85 & 0 & & \\ 0.65 & 0.83 & 0 & \\ 0.13 & 0.71 & 0.79 & 0 \end{bmatrix}$$

- objects 1 and 4 are the most similar
- objects 1 and 2 are the least similar

Cosine Similarity

- A **document** can be represented by thousands of attributes, each recording the *frequency* of a particular word (such as keywords) or phrase in the document.
- Thus, each document is an object represented by what is called a *term-frequency vector*

Document	team coach		hockey	baseball	soccer	penalty	score	win	loss	season
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

- Applications: information retrieval, text document clustering, biologic taxonomy, gene feature mapping, ...
- **Cosine similarity** is a measure of similarity that can be used to compare documents or, say, give a ranking of documents with respect to a given vector of query words.

Cosine Similarity

- Let $\mathbf{x}$ and $\mathbf{y}$ be two vectors for comparison. Using the cosine measure as a similarity function, we have

$$\text{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{||\mathbf{x}|| ||\mathbf{y}||},$$

- where $||\mathbf{x}||$ is the Euclidean norm of vector $\mathbf{x} = (x_1, x_2, \dots, x_p)$ defined as $\sqrt{x_1^2 + x_2^2 + \dots + x_p^2}$.
- $||\mathbf{y}||$ is the Euclidean norm of vector $\mathbf{y}$.
- The measure computes the cosine of the angle between vectors $\mathbf{x}$ and $\mathbf{y}$.
- A cosine value of 0 means that the two vectors are at 90 degrees to each other and have no match.
- The closer the cosine value to 1, the smaller the angle and the greater the match between vectors.

Example: Cosine Similarity

- Ex: Find the **similarity** between documents 1 and 2.

$$\mathbf{X} = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$\mathbf{Y} = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

$$\begin{aligned}x^t \cdot y &= 5 \times 3 + 0 \times 0 + 3 \times 2 + 0 \times 0 + 2 \times 1 + 0 \times 1 + 0 \times 0 + 2 \times 1 \\&\quad + 0 \times 0 + 0 \times 1 = 25\end{aligned}$$

$$||x|| = \sqrt{5^2 + 0^2 + 3^2 + 0^2 + 2^2 + 0^2 + 0^2 + 2^2 + 0^2 + 0^2} = 6.48$$

$$||y|| = \sqrt{3^2 + 0^2 + 2^2 + 0^2 + 1^2 + 1^2 + 0^2 + 1^2 + 0^2 + 1^2} = 4.12$$

$$\text{sim}(x, y) = 0.94$$